Quantification: 682_z23
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	13
interior lacuna count	11
frame-edge lacuna count	2

Interior lacunae only: mean, median and sample SD (n = 11)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1994.91	1519	1026.72	px^2
major_axis_length_px	lacuna shape	85.7455	86.1566	22.5145	px
minor_axis_length_px	lacuna shape	30.4546	26.8577	8.8339	px
aspect_ratio	lacuna shape	2.9668	3.0363	0.8704	unitless
eccentricity	lacuna shape	0.92	0.9442	0.0643	unitless
solidity	lacuna shape	0.8555	0.8762	0.0782	unitless
roots_count	headline	9.7273	9	2.8667	count
ring_length_r30_px	headline	384.091	386	75.5201	px
ring_length_r60_px	reported	949.727	954	165.03	px
owned_length_px	ownership-dependent	2677.02	2005.65	1413.45	px
edge_count	ownership-dependent	125.818	104	71.7841	count
mean_edge_length_px	ownership-dependent	21.8493	21.3532	2.5505	px
perimeter_px	normalised	221.572	210.244	51.1926	px
ring_area_r30_px2	normalised	8909.82	8646	1432.67	px^2
ring_area_r60_px2	normalised	22678.9	22189	3111.45	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	0.9972	1	0.0094	unitless
ring_density_r30	normalised	0.0433863	0.0413851	0.00814286	px^-1
ring_density_r60	normalised	0.0421299	0.0403625	0.00688531	px^-1
roots_per_100px_perimeter	normalised	4.426	4.2807	1.0458	per 100 px
ring_attached_length_r30_px	network v2	345.727	327	74.3923	px
ring_attached_length_r60_px	network v2	753.909	756	184.705	px
ring_length_w_r30_px	network v2	433.334	439.534	82.5219	px
ring_length_w_r60_px	network v2	1072.4	1077.99	179.03	px
sholl_crossings_r10	network v2	9.7273	10	1.9022	count
sholl_crossings_r20	network v2	11.9091	12	1.7581	count
sholl_crossings_r30	network v2	13.2727	14	3.1334	count
loop_count	loops, report only	12.3636	10	10.4525	count
loop_edge_share	loops, report only	0.3372	0.3318	0.1273	share
ring_width_mean_r30_px	width	7.1953	7.2449	0.3824	px

Per field
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measure	value	units
analysed_area_px2	1.02316e+06	px^2
total_skeleton_length_px	37861	px
canalicular_length_density_per_px	0.037004	px^-1
junction_count	877	count
junction_density_per_px2	0.00085715	px^-2
lacuna_count	13	count

lacunae_per_px2	1.271e-05	px^-2
skeleton_component_count	254	count
mean_component_length_px	149.059	px
median_component_length_px	18.5	px
field_length_density_w_per_px	0.0418645	px^-1
field_density_without_flagged_per_px	0.0379094	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	560	px
unattached_component_count	30	count
canalicular_length_density_attached_per_px	0.037004	px^-1
field_length_density_with_unattached_per_px	0.0375513	px^-1
loop_count	180	count
loop_edge_share	0.3609	share
edge_count	1937	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.8184	px
width_median_px	8	px
width_p10_px	4.4721	px
width_p90_px	10.198	px
width_pixels_used	36383	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	510	36.0388	20.5481	1.7539	0.8215	0.8095	-0.2657	13.3863	106.873	True
2	1379	89.4664	21.3431	4.1918	0.9711	0.7971	-1.5436	60.8564	471.593	False
3	1062	83.8665	19.2697	4.3522	0.9732	0.6923	-1.1353	135.124	920.395	False
4	4132	136.42	39.4244	3.4603	0.9573	0.9008	1.5344	431.979	214.138	False
5	1032	65.9485	21.7201	3.0363	0.9442	0.8418	-0.9507	441.636	703.582	False
6	2868	91.0398	40.6809	2.2379	0.8946	0.9166	-1.49	534.719	477.113	False
7	980	60.8458	24.7024	2.4632	0.9139	0.7644	-1.4596	557.819	348.526	False
8	2962	111.965	35.5084	3.1532	0.9484	0.917	1.4275	591.514	47.6516	True
9	2810	108.692	33.8251	3.2134	0.9503	0.9201	1.4475	714.117	172.09	False
10	1297	66.0375	26.8577	2.4588	0.9136	0.9134	-1.189	725.743	851.213	False
11	1519	86.1566	25.1812	3.4215	0.9563	0.8388	1.3918	837.228	113.164	False
12	2083	63.4325	42.3049	1.4994	0.7451	0.9485	-1.2565	905.171	752.206	False
13	2782	91.2948	39.6911	2.3001	0.9005	0.8762	-1.4405	919.61	300.233	False

Network per lacuna

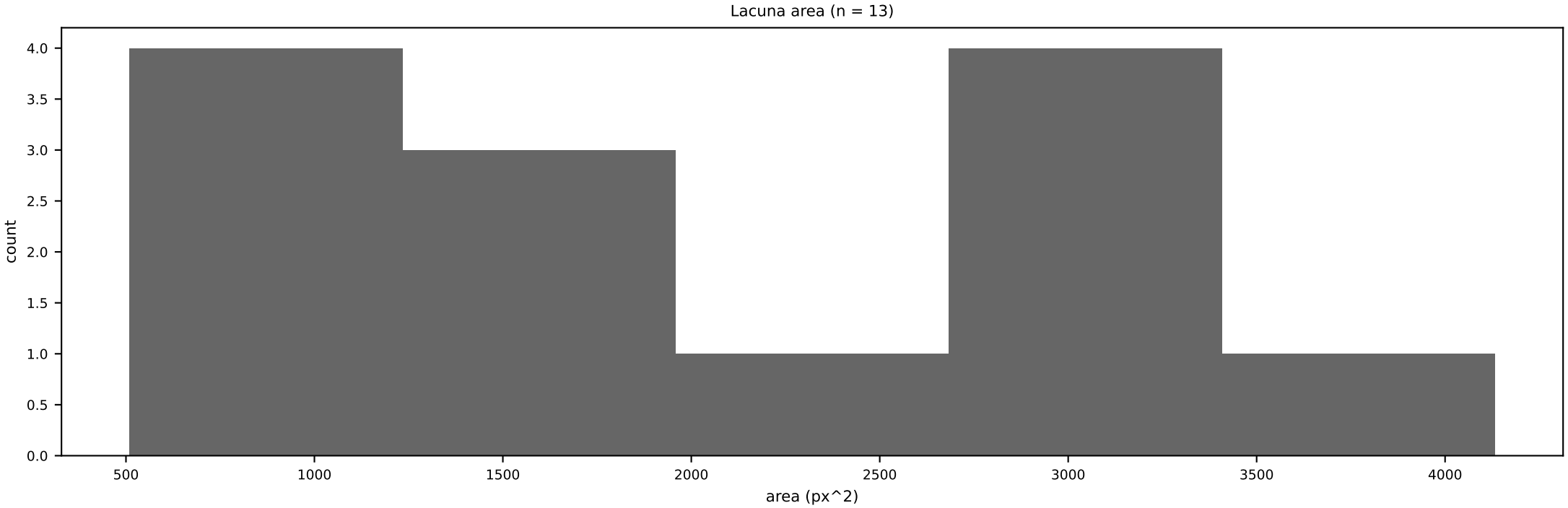
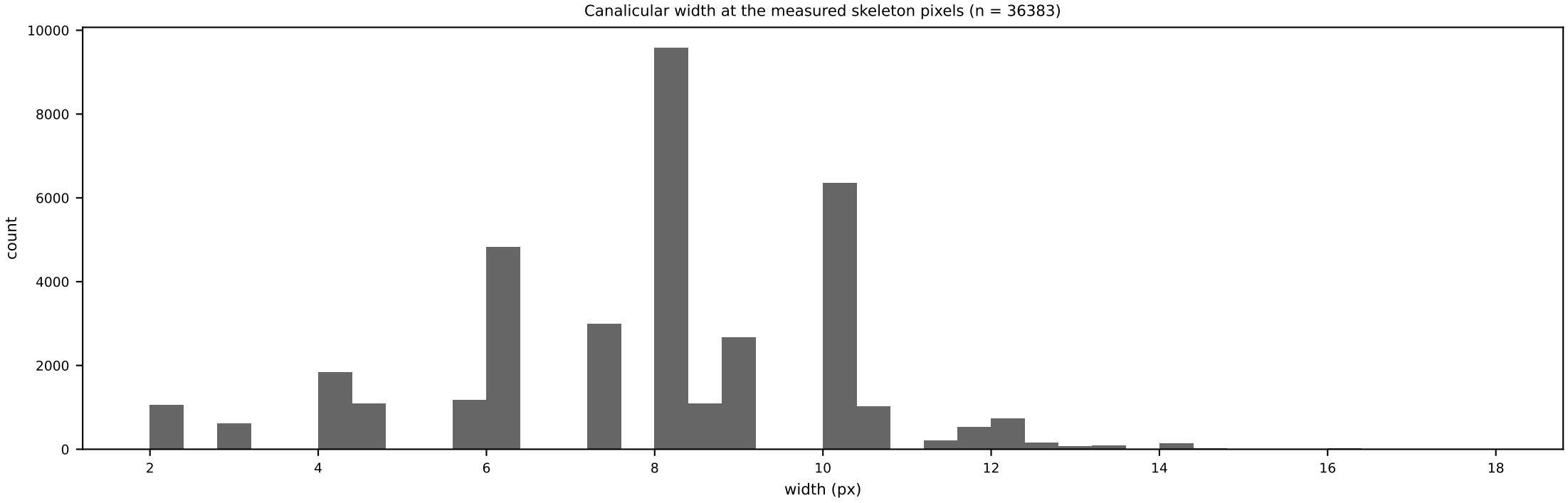
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	2	107	231	2003.95	97	20.6593	
2	8	386	840	2005.65	104	19.2851	7.2461
3	9	440	1067	5400.07	271	19.9264	6.9019
4	15	470	1116	4501.74	217	20.7453	7.6645
5	6	298	790	2560.8	122	20.9902	7.8485
6	10	392	925	4122.75	192	21.4727	7.2449

7	10	460	980	1776.43	92	19.309	7.3791
8	10	351	815	1273.48	53	24.0279	7.4606
9	8	471	1274	3081.6	135	22.8266	7.3243
10	7	289	706	1835.23	66	27.8065	6.9399
11	14	385	954	1387.96	65	21.3532	6.4529
12	8	255	793	1453.53	67	21.6944	7.1906
13	12	379	1002	1321.46	53	24.9332	6.9559

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	105.154	3701	10161	0.6475	0.597	0.0289111	0.022734	1.902
2	219.681	8976	22806	1	0.9689	0.0430036	0.0368324	3.6416
3	210.244	8511	22591	1	1	0.0516978	0.0472312	4.2807
4	335.421	12178	29932	1	1	0.0385942	0.0372845	4.472
5	167.238	7485	20582	1	1	0.039813	0.0383831	3.5877
6	243.522	9472	22056	1	1	0.0413851	0.0419387	4.1064
7	167.924	7349	17555	1	1	0.0625935	0.0558246	5.9551
8	282.894	8678	19332	0.813	0.7462	0.0404471	0.0421581	3.5349
9	263.48	10172	24552	1	1	0.0463036	0.0518899	3.0363
10	174.51	7570	20770	1	1	0.038177	0.0339913	4.0112
11	210.167	8646	22189	1	1	0.0445293	0.0429943	6.6614
12	185.823	7980	21610	1	1	0.0319549	0.036696	4.3052
13	259.279	9669	24825	1	1	0.0391974	0.0403625	4.6282

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	92	162	117.083	255.48	4	2	4
2	300	526	428.463	928.068	10	12	14
3	428	1011	504.375	1222.43	13	13	12
4	432	790	518.505	1237.13	10	14	14
5	281	622	346.635	913.193	9	9	8
6	316	626	439.534	1043.36	9	13	12
7	448	951	520.162	1099.67	11	13	15
8	259	547	400.191	919.21	9	12	11
9	416	1062	525.777	1428.7	12	14	19
10	248	572	327.179	815.009	6	11	10
11	327	768	443.961	1077.99	9	11	17
12	253	609	289.693	907.293	8	9	11
13	354	756	422.392	1123.51	10	12	14

682_z23: distributions, pre-validation, px